

WhatchyaDoin? Separating Intention, Action Realization, Causal Provenance, and Awareness

Abstract

People can want to act, know what to do, and remain capable of doing it while the action still does not occur. When action finally begins, it is tempting to treat onset as evidence for a favored internal mechanism, and to treat awareness of acting as evidence about what caused the onset. This paper argues that those substitutions are not licensed. Prior frameworks already decompose intention, action selection and timing, inhibition, and the prospective and retrospective sources of agency. The proposed contribution is narrower: a cross-literature identification architecture that keeps intention evidence, action-specific realization, causal-provenance evidence, and awareness/origin evidence from substituting for one another. Behavioral realization is defined by a predeclared action-specific criterion, while the causal route to that criterion remains a separate empirical question. Changes in realization can occur without a manipulation supplying a new external action identity, but that observation alone cannot establish that latent internal action content remained unchanged. Likewise, explicit awareness may precede, accompany, or follow portions of action, but any downstream causal role requires awareness-specific evidence rather than a generic monitoring effect. The framework does not propose a new mechanism or universal stage model; it specifies what observations do and do not identify when 'I want to do this' becomes 'I am doing this.'

Keywords: intention-behavior gap; action initiation; volition; action awareness; sense of agency; cognitive control; causal inference

1. WhatchyaDoin?

You are reading this sentence. Now: WhatchyaDoin?

The question can make the ongoing activity explicit. A moment ago, reading was already underway; now the activity may be available in a different way: I am reading this. That small demonstration is deliberately modest. It does not show that the preceding reading was unconscious, that consciousness arrived late, or that any particular neural process initiated the action. It shows only that explicitly recognizing an ongoing action is not automatically identical to the event that made the immediately preceding action occur.

The ordinary problem is broader. There are things people want to do, intend to do, and know how to do, yet do not begin. Then, sometimes with no obvious new instruction, the action starts. A person gets out of the chair, opens the document, sends the message, begins the walk, or resumes the paragraph. Once that transition has happened, another question becomes available: when did the person notice what they were doing, and what does that noticing tell us about why the action began?

Psychology already contains strong accounts of the components. Action-control research asks why intentions are or are not enacted (Koole et al., 2023). Implementation-intention research shows how prior planning can configure later cue-responsive action (Gollwitzer & Sheeran, 2006). Volition research separates questions such as whether, what, and when to act from the subjective experience of intending (Haggard, 2008). Research on agency and motor awareness distinguishes movement, feeling of agency, and later judgment (Synofzik et al., 2008). An earlier paper in *New Ideas in Psychology* even framed 'presence' as a proposed bridge from intention to action (Riva et al., 2011).

Several integrative frameworks define the closest novelty boundary. Pacherie (2008) proposed a three-tiered dynamic model of intention linked to action specification and control in order to explain the contents, sources, and temporal course of action phenomenology. Brass and Haggard (2008) decomposed intentional action into decisions about what to do, when to do it, and whether to act; Zapparoli et al. (2017) later found partial neurofunctional fractionation of those components while also showing that the implicated regions are not unique to intentional action. Chambon et al. (2014) organized sense of agency across an intention-action-effect chain and showed that prospective action-selection fluency and retrospective outcome information can make separable contributions to experienced control. These accounts already rule out treating intentional action, intention, or agency as single undifferentiated events.

The present paper therefore does not claim the discovery of those decompositions. Its remaining scientific function is identificational rather than mechanistic: it asks which observation has actually been obtained at each point in the ordinary transition and which causal conclusions remain underdetermined. Pacherie's framework organizes levels of intention, control, and phenomenology; the What-When-Whether model organizes components of intentional decision; and Chambon and colleagues organize cues contributing to agency. The present framework instead makes action-specific behavioral realization an independently declared observable and then asks whether evidence about intention, realized onset, causal route, and awareness is being allowed to substitute across those categories. That is compatible with the prior frameworks but is not their central organizing task.

Table 1*Functional relationship between closest frameworks and the present identification audit*

Framework	Primary scientific function	Relation to the present audit
Koole et al. (2023)	Action Control Theory explains why intentions are or are not enacted under changing self-regulatory conditions.	Owens the intention-enactment problem. It does not centrally make realized onset, unique causal provenance, and awareness one cross-domain evidential audit.
Pacherie (2008)	Integrates hierarchical intentions, action specification and control, and the phenomenology of acting.	Closest integrative threat. The present paper instead predeclares R_A and asks what each observation licenses about causal history.
Brass & Haggard (2008); Zapparoli et al. (2017)	Fractionates what, when, and whether components of intentional action, with partial neurofunctional dissociation.	Owens content versus timing/permisibility. The present audit does not infer fixed latent content from fixed external instructions.
Chambon et al. (2014)	Separates prospective and retrospective contributors to experienced agency.	Owens multifactorial agency. The present audit treats agency or noticing evidence as distinct from evidence about the origin of realized action.
Riva et al. (2011)	Proposes presence as a bridge from intention to action.	Offers a candidate bridge. The present audit remains agnostic about which bridge operates and asks what evidence would identify any candidate route.
Present paper	Applies one identification audit across intention, action-specific realization, causal provenance, and awareness.	No new mechanism is proposed. Paper-level novelty remains OPEN and collapses if a prior framework is shown to perform materially the same function.

Table 1 is a functional comparison, not a priority claim. Paper-level novelty remains open. If prior work is shown to jointly separate intention or capability, action-specific realization, causal provenance, and awareness as distinct evidential relations, make non-identification from action endpoints central, and supply materially the same discriminating test logic, then the present contribution is subsumed. The argument below is written to make that kill condition testable rather than assume novelty from new terminology.

The present paper does not propose another bridge. It takes a narrower identification problem as its target: when 'I want to do this' becomes 'I am doing this,' what has actually become

observable, what remains latent, and what does consciously noticing the action add? The candidate contribution is compositional. Established literatures often use intention, action onset, causal control, and awareness for different scientific jobs. Bringing them around one ordinary transition exposes several invalid substitutions that become easy to make when those jobs are compressed into a single story.

The central claims are correspondingly limited. Wanting or intending is not the same evidence as behavioral realization. Behavioral realization is not evidence for a unique internal causal route. Explicit awareness of acting is not, by definition, evidence that awareness caused the action to begin. And in some established architectures, enactment can change without the system receiving new action-specific content. None of these claims requires a universal mechanism of will, a new hidden action stage, or a claim that consciousness is always early or late.

2. You Can Want To and Still Not Do It

The first separation is the least controversial and the easiest to lose in ordinary explanation: intention is not behavior. People routinely use statements such as 'I meant to,' 'I was going to,' or 'I really did want to' precisely because an intention can exist without the intended action occurring.

Experimental evidence makes the distinction quantitative. Webb and Sheeran (2006) analyzed 47 experimental tests in which interventions changed behavioral intentions and then measured later behavior. A medium-to-large change in intention produced a smaller change in behavior. The result does not imply that intentions are causally irrelevant. It implies that increasing measured intention and producing the intended behavior are not the same intervention outcome. Intention can matter probabilistically without being sufficient for realization.

Implementation intentions make the point from the opposite direction. A goal intention specifies an outcome one intends to achieve; an implementation intention additionally links a specified situation to a response. Across 94 independent tests, implementation intentions showed substantial effects on goal attainment (Gollwitzer & Sheeran, 2006). Part of their value is that action can later become responsive to a preconfigured cue without requiring another full episode of deliberation at the moment of enactment. This is not evidence that implementation intentions are the universal missing step between wanting and doing. It is evidence that what changes enactment can be something other than simply wanting the outcome more.

Action Control Theory makes the same human problem central at a broader scale. Koole et al. (2023) review decades of work on why people differ in their ability to enact intentions under difficult conditions. The literature already owns the proposition that an active intention can coexist with poor enactment. WhatchyaDoin does not rename that result. It uses it to enforce an evidential boundary: observing an intention gives us evidence about intention; observing an action gives us evidence about action. Neither can substitute for the other.

This distinction also prevents a common reversal. Failure to act does not prove that no intention, desire, conflict, or action tendency existed. If intention is not sufficient for action, non-action cannot be used as a simple diagnostic for absent intention. An empirical study may still conclude that intention was weak or absent, but it needs an independent measurement that supports that conclusion.

For a particular action A , let W denote measured wanting or intending relevant to A , and let R_A denote that A has become behaviorally realized according to a criterion declared in advance. The relation can be written compactly: W is not sufficient for R_A . Even if $P(R_A | W)$ is greater

than $P(R_A \mid \text{not-}W)$, the probabilistic relation does not collapse W into R_A . The formal notation adds nothing mysterious; it simply keeps two measurements from being treated as one.

3. You Can Know What To Do and Still Not Start

A stronger version of the ordinary problem appears when the person not only wants to act but also appears to know what the action is and how to perform it. 'You knew what to do' sounds like an explanation until the action still does not begin. Then it becomes clear that action knowledge, performance capability, and self-initiation are not interchangeable.

Parkinsonian movement provides a useful mechanistic precedent because self-initiated and externally triggered movement can dissociate. Jahanshahi et al. (1995) compared self-initiated and externally triggered movements in people with Parkinson's disease and controls, showing different behavioral and neural organization for the two conditions. More recent synthesis of randomized trials likewise shows that external sensory cueing can change gait performance in Parkinson's disease, although effects differ by cue modality and outcome and the clinical evidence is heterogeneous (Stonsaovapak et al., 2026). The important inference is not that external cues universally solve initiation. It is that an organism may retain substantial movement capability while the conditions under which that capability is recruited differ.

The distinction is easy to state in human language: being able to do something is not the same as being able to start doing it under the current conditions. A successfully executed cued movement shows that at least some relevant execution capacity was available. It does not prove that the self-initiated version failed because one specific internal gate was closed. Multiple differences between cued and self-initiated conditions may contribute. The scientific value lies in the dissociation, not in over-reading it.

Action-control research supplies the nonclinical counterpart. An intention can remain active while action stalls; external structure or changes in control conditions can alter whether it is enacted (Koole et al., 2023). Again, the point is not that the action is fully represented in every failed attempt. Some actions genuinely require more planning, information, conflict resolution, or construction. The narrower claim is that missing action-specific knowledge is not the only possible explanation for non-initiation.

This matters for both science and everyday interpretation. If a person later performs the action competently, it is tempting to retrospectively say that they 'could have done it all along.' That statement may also overreach. Present performance does not prove that the earlier state contained every enabling condition required for self-initiation. What the later performance can establish more safely is that the system possessed or acquired sufficient capacity to execute under the later conditions. The causal difference between earlier non-action and later action remains to be identified.

4. Sometimes the Difference Is Not New Content

If wanting can be present without action, and action capability can be available without self-initiation, what changes when the action finally occurs? There is no single answer. The useful constraint is that the answer is not always 'the system received a new description of the action.' Several established literatures provide examples. In implementation intentions, the action and cue-response relation can be specified prospectively, with later behavior becoming more readily triggered when the critical situation is encountered (Gollwitzer & Sheeran, 2006). In the affordance-competition account, multiple potential actions can be specified in parallel while ongoing processes bias competition toward one outcome (Cisek, 2007). In action-control research, the same intention can be more or less likely to govern behavior as self-regulatory

conditions change (Koole et al., 2023). These theories are not identical and should not be merged into a single mechanism. They share only a structural possibility: action-relevant organization may exist before the conditions that make it behaviorally effective.

Brass and Haggard's (2008) What-When-Whether model is especially important here because it already separates action content from the timing and permissibility of action, and Zapparoli et al. (2017) provide meta-analytic support for partial independence among those components.

WhatchyaDoin does not recast that separation as a new content-versus-transition mechanism. Its narrower use of the precedent is evidential: a measured change in when or whether an action occurs does not, by itself, identify what changed in the internal action representation, just as a fixed external instruction does not prove that internal content remained fixed.

Three claims must therefore be kept separate. First, a manipulation may add no new external instruction or action identity: the person is still being asked to perform the same action. Second, the person's latent internal action-specific representation may or may not change through retrieval, replanning, recombination, attentional selection, or policy updating. Third, it would be stronger still to claim that nothing internally changed. Only the first claim can usually be secured from the task instructions alone. The second requires independent representational evidence, and the third is not the intended claim: if realization changes, some causally relevant relation or condition must differ even when the action identity does not.

That possibility supports an important correction to ordinary language. Sometimes nothing new has to be explained to you. The relevant difference is whether what was already available becomes effective in what happens next. 'Effective' here is descriptive of a causal relation, not the name of a hidden stage. The paper does not insert an unobserved box called effective control

between intention and behavior and then infer that box whenever behavior occurs. Doing so would simply rename the transition rather than explain it.

The distinction can be kept clean by separating content from causal contribution. Suppose W is an intention relevant to A and R_A is the measured realization of A . The question 'Did W cause or contribute to R_A ?' concerns the relation $W \rightarrow R_A$. It is not answered by adding another mandatory state between W and R_A . Evidence for the causal relation must come from intervention, temporal structure, mediation, comparative conditions, or another design capable of distinguishing alternatives.

This is also where the present paper differs from proposals that identify a specific missing bridge. Riva et al. (2011), for example, proposed presence as a link between intention and action. That is a substantive theoretical proposal. The present argument is deliberately agnostic at that level. It asks what any candidate bridge would have to explain and what evidence would distinguish it from alternatives. A cue, a control-state change, conflict resolution, a learned trigger, altered motivation, or a neuromodulatory event may all matter in different cases. The common identification rule survives even if no single one is universal.

5. A Trigger Does Not Have to Contain the Action

A compact systems analogy helps make the preceding point intuitive. In a computer, an interrupt request does not need to contain the handler that eventually runs. The request can change whether already-existing machinery receives control. Brains are not operating systems, and the analogy should stop there. Its only job is to separate a transition-enabling signal from the action-specific organization whose expression changes.

Dopamine provides a concrete biological proof of principle. In mice, da Silva et al. (2018) found transient substantia nigra dopamine-neuron activity before self-paced movement initiation. The

activity was related to the probability and vigor of future movement but was not action-specific. Brief inhibition while animals were immobile reduced subsequent movement initiation; brief activation increased it. Manipulations after movement initiation had little effect on ongoing movement in the tested conditions. The result does not establish dopamine as a universal action command or as the decision to act. It shows that a causal signal can alter whether movement begins without uniquely specifying which movement is represented.

Noradrenergic theory supplies another architecture. Bouret and Sara (2005) proposed that phasic locus-coeruleus activity can facilitate reconfiguration of target networks during cognitive shifts. On that account, the behavioral outcome depends substantially on the target network and current state; the neuromodulatory event is not a replacement behavior encoded in miniature. Whether this specific account explains any given ordinary action is an empirical question. The relevant proof of principle is simply that transition-related modulation and action identity can be different informational roles.

The affordance-competition framework makes the same structural distinction without requiring a neuromodulatory story. Cisek (2007) describes action specification and selection as concurrent processes in which multiple possibilities may be represented and biased over time. Again, biasing a competition is not the same thing as supplying the identity of every candidate being biased.

These examples are intentionally heterogeneous. They should not be welded into a single pathway. Their value is logical and empirical: real systems exist in which a signal or control process changes the probability, vigor, selection, or organization of action without itself constituting a complete action specification. Therefore, when an action begins after a period of

non-action, it is not scientifically necessary to assume that a new action-specific thought or command appeared at the moment of onset. Sometimes it may have. Sometimes it may not have.

6. Starting Tells Us Less Than We Think

The word 'doing' can create a hidden circularity if it is defined by whatever internal control process a theory wants to explain. The safer move is operational. For each action class, declare what counts as behavioral realization before using realization as evidence.

For reaching, R_A might be a kinematic movement-onset threshold. For standing, it might be a specified postural transition. For typing, it might be the first task-directed keystroke under a predeclared rule. For sending a message, the scientifically relevant onset could be composition, the send event, or another criterion, depending on the question. For reading, eye motion alone is inadequate because eyes move for many reasons; a study might instead define sustained text-directed sampling plus a comprehension criterion. There is no need for one universal physical marker across all action classes.

Once R_A is defined independently, a central inference rule becomes visible: R_A does not identify X in $X \rightarrow R_A$. The action started tells us that something changed. It does not by itself tell us what made it change.

This is an observational-equivalence problem. The same outward onset may be compatible with deliberate prospective planning, cue-triggered initiation, practiced control, resolution of conflict, a change in motivational state, external support, altered neuromodulatory conditions, or a combination of these. An endpoint cannot choose among latent organizations that predict the same endpoint. Additional measurements or interventions are required.

6.1 A worked observational-equivalence case: beginning to type

Consider a simple typing task. Before each trial, a participant is shown and asked to remember the same short phrase. The phrase, keyboard, posture, and instructions remain unchanged. The participant is then given a waiting period and told to begin typing the phrase when ready. Define behavioral realization in advance: R_A occurs at the first task-directed keystroke, or, more conservatively, at the first short sequence that matches the target strongly enough to exclude an accidental keypress. Once that criterion is crossed, the experiment has established something concrete: this instance of the typing action became behaviorally realized at that time. It has not yet established why.

Several latent organizations can produce the same observed first keystroke. A prospective plan may already have been sufficiently specified and become behaviorally effective as urgency or readiness changed. A learned cue or contextual feature may recruit a preconfigured response.

The participant may retrieve or reconstruct part of the phrase immediately before typing.

Attention may shift toward the task, competition with another activity may resolve, or a control state may change while the externally specified action remains identical. These routes need not produce different visible onset events. If each yields the same first correct keystroke at the same latency, then R_A alone leaves them observationally equivalent. Even a clean record of the subsequent correct phrase establishes successful execution under the realized condition, not the unique causal history of the transition.

The audit changes what the study must measure next. Randomizing a transition-supporting cue can show that the cue changes onset probability or latency without supplying a different external action identity. Independent probes or assays of remembered action identity can constrain some retrieval or replanning explanations, although the probes themselves may perturb the process.

Measures of motor preparation, task engagement, or competing-action state can further separate candidate routes when they are theoretically justified. But a null difference on one representational measure does not prove that every latent representation was constant. The strongest licensed conclusion is always tied to what was independently manipulated and measured: the endpoint identifies realization; additional evidence progressively narrows causal provenance.

The rule prevents several familiar errors. Temporal precedence of an intention does not by itself establish that the intention was the sole causal authority. Successful onset does not establish the presence of a particular gate. Non-action does not establish the absence of intention. A coherent later report does not reconstruct the unique internal route that produced the action. Each of these may be supported by additional evidence, but none follows from onset alone.

The same discipline applies to the phrase 'self-initiated.' In experimental work, self-initiated often means that onset was not locked to an immediate external imperative. It does not mean causeless, unconditioned, or generated by a single internal source. A self-paced movement can still depend on prior instructions, task context, learned policies, internal state, environmental affordances, and ongoing neural dynamics. The label describes the experimental arrangement more safely than it identifies the complete causal provenance of the act.

The practical payoff is restraint. Instead of asking the action to disclose its own explanation, we can treat realization as one measurement and design the rest of the experiment around what remains uncertain.

7. Then You Notice

Only after intention, realization, and causal provenance are separated does action awareness become easier to discuss. Noticing yourself doing something is another fact about the episode. It is not automatically the event that made the action begin.

The classic literature on the timing of conscious intention encouraged a simple temporal story in which neural preparation comes first and consciousness learns about the decision later. That story is no longer licensed as a universal interpretation. Matsushashi and Hallett (2008), using real-time probes rather than only retrospective clock reports, estimated conscious intention substantially earlier than traditional Libet-style reports. Banks and Isham (2009) showed that reported decision time can shift with manipulated feedback about movement timing, demonstrating that retrospective timing reports can incorporate post-action information. Schurger et al. (2012) showed that the readiness potential can emerge from accumulation of spontaneous fluctuations toward a threshold, undermining the inference that the averaged potential is itself an already-made unconscious decision. Neafsey (2021) reviews the resulting problems for strong conclusions drawn from readiness-potential and clock paradigms.

These findings do not prove the opposite universal - that conscious intention always causes action before neural preparation. They remove the license for a fixed 'unconscious decision first, consciousness later' pipeline. Different tasks, measures, and operational definitions capture different relations.

Motor-awareness evidence makes the separation more direct. During neurosurgical stimulation, Desmurget et al. (2009) reported that parietal stimulation could evoke an intention or belief that movement had occurred without corresponding electromyographic movement, whereas premotor stimulation could evoke overt movement that participants denied having made. These unusual

stimulation conditions are not a model of ordinary daily action, but they demonstrate that movement, intention, and awareness are dissociable phenomena rather than one indivisible event. Sense-of-agency research adds another layer. Synofzik et al. (2008) distinguish a lower-level feeling of agency from a higher-level judgment of agency and argue that multiple cues can contribute to each. A person's later answer to 'Did you do that?' therefore need not be a direct readout of the entire causal history. It is itself a psychological outcome constructed from available information.

Chambon et al. (2014) make this non-unitary structure explicit along the intention-action-effect chain. Their review distinguishes prospective contributions to agency from action-selection fluency and retrospective contributions from action outcomes, including experiments in which these sources can be varied independently. That work already owns the claim that agency judgments can be built from information available at different times. The additional constraint used here is different: neither an agency judgment nor a report of noticing is treated as a transparent record of the causal provenance of behavioral realization.

The word awareness therefore cannot carry all of these constructs at once. Conscious intention concerns awareness of an intention, urge, or prospective decision to act; motor awareness concerns awareness that movement is occurring or has occurred; explicit noticing is the metacognitive recognition that one is presently engaged in a particular activity; an online report is a contemporaneous behavioral response to a probe; a retrospective report is a later reconstruction from memory and other cues; a feeling of agency is an experience of authorship or control; and a judgment of agency is an explicit attribution that may integrate additional evidence. Pacherie (2008), Synofzik et al. (2008), and Chambon et al. (2014) provide different

reasons not to collapse these phenomena. A timing result for one construct is not automatically a timing result for the others.

For the present framework, N_A therefore denotes only an operationally specified awareness measure relevant to action A, not a universal state called consciousness. A study using a probe must state what the probe asks, when it occurs, what response counts as awareness, and how the probe itself may alter attention or action. A study using retrospective timing must treat the report as a reconstruction-sensitive measurement rather than transparent access to first phenomenal occurrence. A failure to report under one method establishes absence of that report under that method; it does not, without additional evidence, establish absence of phenomenal presence, action-related information, or every form of awareness. The minimal claim remains that R_A is not identical by definition to N_A. Depending on task and measure, N_A may precede R_A, accompany it, follow part of it, or fail to be captured.

8. Noticing May Still Change What Happens Next

Separating awareness from action onset does not imply that awareness is epiphenomenal. It changes the causal question. The fact that noticing need not have originated an action already underway leaves open whether explicit awareness later becomes information that changes continuation, correction, stopping, redirection, or learning. That downstream role is a hypothesis requiring its own causal evidence, not a consequence of the separation itself.

Return to reading. The question 'WhatchyaDoin?' may make the current activity explicit. A reader might then slow down, reread a sentence, stop, search for a related source, or deliberately continue. Those observations make the downstream question plausible, but they do not isolate awareness as the cause: the prompt also changes attention, expectancy, task set, and response strategy.

Performance-monitoring research shows that behavior can change after errors through post-error slowing, altered interference control, and changes in later accuracy (Danielmeier & Ullsperger, 2011). That literature does not, by itself, establish that explicit conscious noticing caused those adjustments; some adjustments occur without reportable awareness. It therefore supplies a useful negative control on the inference: later behavioral change is not sufficient evidence for awareness-specific causation. The narrower proposition needed here is only that separating awareness from action initiation does not license the opposite universal claim that explicit awareness can never influence later behavior.

This asymmetry avoids two opposite overclaims. One treats coherent awareness as proof that consciousness initiated the action. The other treats evidence of pre-awareness processing as proof that consciousness can never matter causally. Neither follows. A local analysis instead asks: awareness of what, measured how, relative to which action boundary, manipulated or contrasted against what controls, and with what downstream consequence?

The answer may vary across action classes. Explicitly noticing a typing error, fatigue, or a wrong route could in principle contribute to correction or redirection; in other cases, control can adjust without reportable awareness. The framework does not privilege either outcome. It requires an awareness-specific causal claim to be supported by evidence that distinguishes awareness from correlated changes in attention, feedback processing, expectancy, caution, or task set.

9. What Would Distinguish the Explanations?

A useful conceptual distinction should change experiments. The present framework yields five families of discriminating tests. They are not one grand experiment, and no single result would establish a universal action architecture.

9.1 Intention versus realization

Measure intention independently, hold it approximately stable where feasible, and manipulate an established action-support condition. The support might be cueing, implementation planning, urgency, environmental structure, or another justified variable. Measure realization with an action-specific criterion. A result in which R_A changes substantially while W changes little would directly show why intention and realization should remain separate measurements.

9.2 Capability versus self-initiation

Compare self-initiated and externally supported versions of comparable actions while measuring post-onset execution separately from onset. The most informative pattern is not simply better cued performance; it is a larger effect on initiation than on execution capability. Parkinsonian cueing paradigms provide one precedent, but the logic can be applied more broadly when the action class permits it.

9.3 Content versus transition conditions

A concrete version can reuse the typing case. Hold the target phrase, keyboard mapping, instructions, and measured pretrial knowledge as constant as the design allows. Manipulate a transition-relevant condition - for example, a cue, urgency signal, contextual support, or theoretically justified control manipulation - that does not itself tell the participant which phrase to type. Independently record R_A as onset probability or latency and record execution quality after onset. This design can establish whether the manipulation changes enactment without supplying a new external action identity.

The result does not establish that internal action-specific content was unchanged. The participant may retrieve the phrase again, replan its initial segment, shift attention toward a stored plan, change a policy for when to release the response, or resolve competition with another action.

Those latent changes can remain observationally equivalent while the external instruction is perfectly constant. A stable accuracy score after onset is also insufficient by itself: successful execution can coexist with different pre-onset representational histories.

A stronger content-versus-transition inference therefore requires evidence aimed at the competing internal-content account. Relevant strategies include predeclared representational assays, restricted manipulations chosen because they should alter transition support while leaving specified content measures invariant, repeated or intermittent probes whose reactivity is modeled, and explicit comparison of causal models that assign the observed onset effect to content change versus transition conditions. If the available measures cannot distinguish those models, the correct conclusion stays weak: the manipulation changed realization without providing a new external action identity. That weaker result is still informative, but it does not license the claim that no new internal content was formed.

9.4 Realization versus awareness

Define R_A independently of N_A . Measure awareness prospectively or with probes whose reconstruction limits are known rather than assuming a retrospective timestamp is transparent. The critical question is whether the two relations can vary independently. Dissociation need not mean total independence; even partial separability is enough to reject identity by definition.

9.5 Awareness as downstream causal input

The downstream-awareness question is harder because an intervention intended to increase noticing usually changes other things at the same time. In the typing case, a prompt such as 'notice what you are typing' may increase explicit recognition, but it also changes attention, expectancy, task demand, caution, response strategy, and perhaps feedback processing. If typing

subsequently slows or becomes more accurate, that result identifies an effect of the broader monitoring condition, not an awareness-specific causal effect.

A stronger test begins only after R_A has been crossed and requires a locally defined N_A. The design should include an awareness-targeted manipulation or contrast, a validated trial-level awareness measure, and comparison conditions matched as closely as possible for sensory stimulation, response requirement, feedback, attention, expectancy, and task demand.

Downstream outcomes can then include continuation, stopping, correction, redirection, or later learning. The critical discriminating pattern is not merely that a monitoring prompt changes behavior, but that variation attributable to the specified awareness construct predicts or mediates the later effect while the principal alternative pathways are independently constrained.

Even this does not guarantee complete identification. Awareness may be inseparable from some attentional or metacognitive changes in a given paradigm, and a report used to validate N_A may itself alter the action. The framework therefore permits a negative result at the level of identification: if awareness cannot be varied or measured without changing the competing processes that could produce the same downstream behavior, awareness-specific causation remains OPEN in that design. The scientifically justified statement is then narrower - a monitoring or evaluation condition changed what happened next - rather than the stronger claim that conscious noticing was the causal mediator.

Across all five test families, the important methodological rule is the same: do not let the dependent variable smuggle in the causal explanation. If 'successful initiation' is defined as 'the intention gained control,' then observing successful initiation cannot independently establish that the intention gained control. If 'awareness' is defined by a later verbal report, absence of that

report cannot automatically establish absence of every form of phenomenal or action-related awareness. Measurements should earn the inferences attached to them.

The framework also supplies failure conditions for itself. If an action class cannot meaningfully separate intention, realization, and awareness, then the distinctions should not be forced onto it.

If manipulating candidate transition conditions changes measured or plausible latent action-specific content in lockstep, the content-versus-transition claim is not identified in that paradigm.

If awareness-targeted manipulations cannot be separated from attention, expectancy, feedback, task demand, or strategy changes, awareness-specific downstream causation is likewise not

identified. If awareness and realization cannot be measured with sufficient temporal or conceptual independence, strong ordering claims should be withheld. The architecture survives only where the measurements make different predictions or support different interventions.

For practical use, the framework can be compressed into five questions applied in order. What was actually observed? What relevant construct was measured independently rather than inferred from that observation? What was manipulated? Which latent explanations still predict the same result? Which conclusion does the evidence discriminate, and which stronger conclusions remain unavailable? The value of the audit is not that every study must measure everything. It is that each study should stop its inference at the boundary its measurements and interventions have actually crossed.

10. What the Ordinary Story Gets Wrong

Ordinary explanations often compress a sequence into one causal sentence: I wanted to, so I decided to, then my brain started it, then I noticed. That story can be true for a particular episode, but the endpoint does not make it true by itself.

The first substitution is intention evidence for realization evidence. A person can report genuine intention without the action occurring, and experimentally increasing intention does not produce a one-for-one change in behavior (Webb & Sheeran, 2006).

The second substitution is realization evidence for causal-provenance evidence. Once the action begins, we have a new observable fact: R_A. But action onset is compatible with multiple latent organizations. A cue, prior plan, competition among candidates, self-regulatory shift, or neuromodulatory change may contribute. The observed action does not choose among them.

The third substitution is action-awareness evidence for evidence that awareness caused realization. Awareness can be early in some tasks, reconstructed in others, dissociable from movement in unusual stimulation conditions, and informed by prospective and retrospective cues (Banks & Isham, 2009; Desmurget et al., 2009; Matsushashi & Hallett, 2008; Synofzik et al., 2008). Therefore, the fact that a person can report the action does not identify awareness as its origin.

The fourth substitution is 'no new action-specific content' for 'no causal change.' This is especially important. If a transition-enabling signal does not specify the action, it can still matter causally. The da Silva et al. (2018) result is a clean example: manipulating a non-action-specific dopamine signal changed the probability and vigor of future movement. The absence of new action identity does not mean nothing changed; it means the causal change may have occurred at another level of organization.

These corrections produce a simple chain of evidential discipline: intention evidence is not realization evidence; realization evidence is not causal-provenance evidence; awareness evidence is not origin evidence. Each relation can be scientifically meaningful. The mistake is allowing one to stand in for another without additional evidence.

The functional comparison in Table 1 shows why these corrections are not claims to have discovered the component distinctions. Action-control theory, What-When-Whether fractionation, action phenomenology, prospective and retrospective agency accounts, and presence-based bridging proposals already own substantial parts of the territory. The present composition adds no competing stage model. Its proposed use is narrower: place the same evidential audit across neighboring literatures so that intention measures do not establish realization, realization does not identify causal provenance, and awareness or agency measures do not establish the origin of the realized action. This is the paper-level claim that remains OPEN rather than assumed novel.

11. Limits

The framework is intentionally narrower than a general theory of action. It is aimed at ordinary goal-related actions for which intention, action knowledge or capability, behavioral realization, and awareness can be meaningfully discussed or measured. Reflexes, seizures, autonomic responses, externally forced movements, compulsive acts, highly automatized skills, novel deliberative construction, and covert cognitive actions may require different boundaries or different realization criteria.

Reading is illustrative, not universal. A person can move their eyes across text without comprehending it, and comprehension can continue across pauses in eye movement. Any experiment using reading must specify what counts as realization for the question being asked.

The opening example is therefore a phenomenological prompt, not a measurement protocol.

The neurobiological examples are proofs of architectural possibility, not a proposed universal mechanism. Dopamine is not the decision, norepinephrine is not a generic interrupt request, and neither is claimed to be necessary for every action. Clinical dissociations do not prove that the

same causal organization governs ordinary action in healthy participants. The paper uses them only where they establish that execution capability, initiation conditions, action specification, and transition-related signals can come apart.

The paper also does not settle whether conscious intention can be causally efficacious in any particular action. The empirical literature does not support a universal claim that consciousness is always late, but rejecting that universal does not establish the opposite. The framework requires local causal attribution: which awareness measure, which action, which time boundary, and which intervention?

Finally, the paper-level contribution is compositional rather than component-level. Action Control Theory and implementation intentions already own intention-enactment distinctions; Pacherie (2008) integrates levels of intention, action control, and phenomenology; Brass and Haggard (2008) and Zapparoli et al. (2017) own the What-When-Whether decomposition; volition, affordance competition, basal-ganglia and neuromodulatory work already own major initiation and selection mechanisms; and Chambon et al. (2014) and related agency work already own prospective/retrospective contributions to experienced agency. The proposed value is narrower: applying one cross-literature identification audit to the ordinary transition from intending, to observed realization, to claims about causal route, to awareness. That value should contract to SUBSUMED if a prior framework is shown to perform materially the same evidential function at the same level.

12. WhatchyaDoin?

There are moments when wanting is not the missing piece. Knowledge is not obviously the missing piece. Ability may not be the missing piece either. Then the action starts.

That onset is real and measurable, but it does not disclose its own cause. Established research provides multiple ways in which an available action can become more likely to occur: prior planning, cueing, competition and biasing, changes in self-regulatory conditions, and neuromodulatory changes among them. None needs to be universal for the identification rule to hold.

Their shared lesson is narrower: what is available to a system is not necessarily what currently governs what the system does. A transition-enabling influence need not contain the action it helps realize. And once realization occurs, the fact of onset remains different from an explanation of onset.

Awareness is another relation again. It may precede, accompany, or follow portions of action depending on task and measurement. Whether explicit awareness then changes what happens next is a separate causal question requiring awareness-specific evidence. Therefore wanting, knowing, starting, and noticing should not substitute for one another as evidence.

Return to the title. WhatchyaDoin? The question can make an ongoing action explicit, but it does not reveal why that action began. The scientific question is not only what you wanted or what you noticed yourself doing. It is: what became true when this became the thing you were actually doing, and what evidence would let us know why?

Data availability

No new data were created or analyzed in this theoretical article. Data sharing is not applicable.

Ethics statement

This theoretical article did not involve human participants, animals, or the collection of new empirical data; ethics approval and informed consent were not applicable.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used OpenAI ChatGPT to assist with research synthesis, adversarial critique, organization, drafting, and revision. The author independently evaluated the scientific claims, verified the cited sources and bibliographic information, reviewed and edited the generated material as needed, and takes full responsibility for the content of the published article.

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